CulinarySense — Voice-Based Recipe Guidance using ASR + RAG + LLMs

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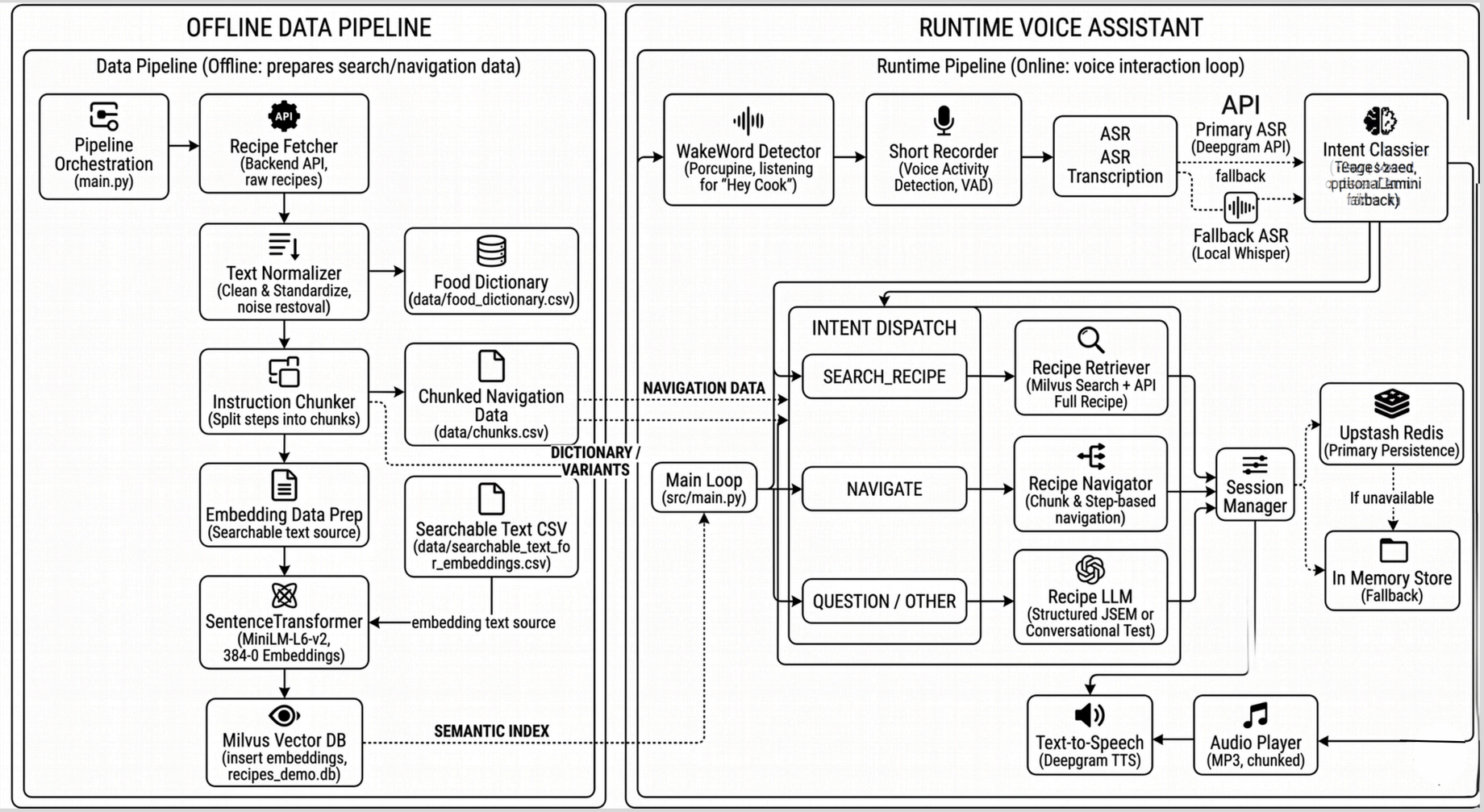
Motivation

- Cooking requires hands-free interaction; phones/screens are inconvenient in kitchens.
- **Goal**: Build a voice-controlled recipe assistant that listens, understands, retrieves recipes, and guides step-by-step.

System Overview

- **Wake Word Activation:** “Hey Cook” using Picovoice Porcupine
- **Speech Recognition (ASR):** Deepgram API (primary) with Whisper (local fallback)
- **Voice Activity Detection:** Short recorder with VAD for efficient audio capture
- **Text Processing:** Normalization + domain-specific correction (food dictionary + fuzzy matching)
- **Retrieval:** SentenceTransformer embeddings → Milvus Vector Database
- **Recipe Access:** Backend API fetches full ingredients and step-by-step instructions
- **Generation:** LLM (Gemini 2.5 Flash) for structured or conversational responses
- **Speech Output:** Deepgram TTS (chunked audio playback)
- **Session Management:** Redis (primary) with in-memory fallback



Data Flow

- User says “Hey Cook” → audio recorded
- Whisper → text → corrected → vectorized
- Milvus returns top recipe
- API fetches full details

Key Features

- Real-time ASR + RAG pipeline
- Domain-specific correction for food vocabulary
- Fast response using Whisper-small + Gemini Flash
- Text → structured prompt → conversational cooking steps

Conclusion

- **Hands-free Cooking Assistant:** A voice-based system using Whisper ASR, Milvus vector search, and Gemini LLM/TTS to retrieve recipes and guide users step-by-step without touching a screen.
- **Intelligent Dialogue System:** Includes intent classification, session tracking, and navigation logic (next/prev/repeat) .